A FIELD GUIDE FOR FACTORIES & LABS

Why Antimicrobial Test Results Vary

Same product, same fabric, very different numbers. The spread is almost never the chemistry — it's the test setup. Five variables move the result, and any factory or lab changing one of them reintroduces the swing. Here's what each one does, and the two ways to make results line up.

THE FIVE VARIABLES THAT MOVE THE NUMBER

Variable	What it does to the result	FUZE-correct setting
Test method	AATCC 100 / JIS L1902 are **absorption** tests — bacteria pressed into stacked humid fabric. They read high for almost any finish and were built for ion-leaching chemistries. ASTM E2149 is a **dynamic-contact** test — the method designed for non-leaching, contact-kill chemistries like FUZE. A lower absolute % on E2149 is normal, not weaker product.	**ASTM E2149 as the primary method; AATCC 100 / JIS only at F1 as collaborating data**
Sterilization	The single biggest swing. Autoclave (120°C) opens synthetic fibers and the surface-bound metamaterial sinks in → near-zero kill. UV preserves the surface chemistry.	**UV sterilization only — never autoclave**
Growth broth	High-sulfur broths quench the chemistry and depress the number; low-sulfur media let it work.	**Low-sulfur Mueller Hinton broth**
Contact time	The 1-hour default is calibrated for fast ion-leachers. FUZE is contact-kill with slower kinetics and needs the full window.	**24-hour contact time minimum**
Organism	*Klebsiella* is non-motile and encapsulated — it physically can't reach the surface chemistry, so it always reads lowest. Motile *E. coli* / *Staph* find it and read higher.	**E. coli ATCC 25922 primary; Staph / Kleb as specified**
Loading (ICP)	Without ICP you're comparing fabrics, not product — dose varies by substrate, pickup and density. One mill ran ~2× the dose and still read lower, because the parameters above dominate, not the dose.	**ICP-confirm mg/kg first; F1 (1.0 mg/kg) baseline**

TWO WAYS TO END THE VARIANCE

OPTION 1 — Enforce one protocol

Lock every factory and every lab to the settings above — same method, sterilization, broth, contact time and organism — and enforce it on every run. Technically perfect, but hard to police across ten mills and several labs; one drift brings the swing back.

OPTION 2 — Standardize the dose (recommended)

If we're not going to police every factory and lab, put more product on: start every mill at **F1 (1.0 mg/kg)**. The higher loading overwhelms the lab-to-lab and chemistry differences and passes on every method and organism. Approve the whole program, then optimize the dose down per fabric once the baseline is locked.

THREE TRUTHS THAT SETTLE MOST ARGUMENTS

- **Non-leaching means durability, not activation.** Leaching antimicrobials need washing to expose fresh active, so their numbers can climb. FUZE is bonded into the fiber — the win is the **same** result at wash 0 and after 15 / 25 / 50 / 100. Flat is good.
- **A lower ASTM % is not weaker product.** It's a harder, mechanism-matched test. Absorption tests (AATCC / JIS) flatter almost any finish; E2149 is the honest contact-kill measure.
- **90%+ reduction is complete odor control.** A passing fabric holds ~150,000 organisms that would otherwise grow past 7 trillion in 24 hours. That is total odor and microbial control — separate from a true "sanitizer" (<10 CFU) claim.

The proof it's test prep, not chemistry: the same fabric, sterilized by autoclave vs. UV, swings from 0–87% to 99.9% reduction. Nothing about the product changed — only how the lab prepared the sample.